
PREDICTION OF EPILEPTIC SEIZURES USING EEG SIGNALS

Signal and Systems Project - Phase 1



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Contents

1	Abstract	3
2	Electroencephalography (EEG)	4
2.1	Electrode's Naming and Placement	4
2.2	Frequency Bands of EEG	5
2.3	Sampling Frequency and Nyquist Criterion	5
3	Preprocessing Data	7
3.1	Why Preprocessing?	7
3.2	Initializing EEGLAB	7
3.3	Applying Filter	7
3.4	Re-Referencing	10
3.5	Independent Component Analysis (ICA)	11
3.6	Source Localization	12
3.7	Analysis of Detected Components	12
3.8	Epoching and Reshaping	15
3.9	Subsampling	16

1 Abstract

The prediction of epileptic seizures holds critical importance due to its potential to markedly improve the quality of life for individuals afflicted with epilepsy. Early seizure prediction can provide timely alerts, enabling preemptive interventions that prevent injuries and mitigate the anxiety associated with the unpredictability of seizures. Furthermore, it facilitates the development of responsive therapeutic interventions, such as neuromodulation techniques, thereby enhancing patient outcomes.

The application of signal processing techniques to electroencephalogram (EEG) data is fundamental for effective seizure prediction. EEG recordings capture the brain's electrical activity, offering extensive information on neural dynamics. Advanced signal processing methodologies, including time-frequency analysis, machine learning, and deep learning algorithms, can discern subtle patterns and precursors to seizures that are not detectable through conventional analysis. By extracting pertinent features and enhancing signal quality, these techniques significantly improve the accuracy and reliability of seizure prediction models. Therefore, leveraging EEG signal processing is indispensable for developing robust, real-time seizure prediction systems, presenting a promising approach for advancing epilepsy management and patient care.

In Phase 1 of this project, we aim to establish the theoretical foundations and perform initial data analysis necessary for the prediction of epileptic seizures using spectral entropy-based features of EEG signals. This phase involves comprehensively understanding and preparing the EEG dataset, including data loading, exploration, and preprocessing. Specifically, We will implement preprocessing certain steps to clean the data and extract meaningful features.

This foundational work is crucial for ensuring the integrity and quality of the data, setting the stage for the advanced analysis and machine learning tasks in the subsequent phase of the project.

2 Electroencephalography (EEG)

2.1 Electrode's Naming and Placement

Question 2.1. what does each electrode's name stand for? Explain the naming method used in the 10-20 EEG system.

The 10-20 EEG system depicted in Figure 1 illustrates the standardized method for electrode placement, which is used to ensure reliable and reproducible recording of electrical activity in the brain. This system is widely adopted in clinical and research settings due to its reproducibility and anatomical relevance. The naming convention for each electrode in this system is methodically structured, reflecting both the location and the underlying cortical area.

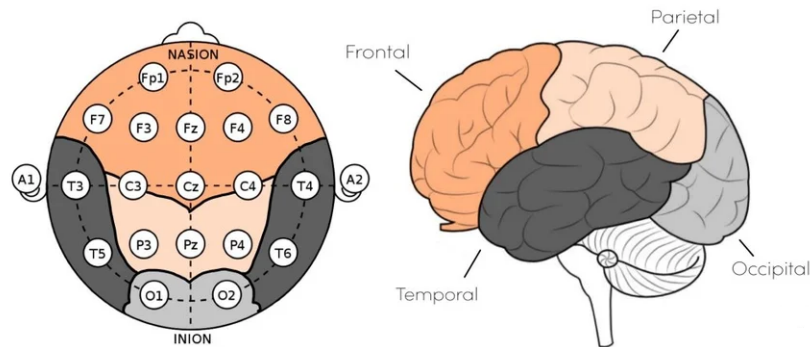


Figure 1: EEG 10-20 Electrode Placement System

Naming Convention

The electrodes in the 10-20 system are labeled with a combination of letters and numbers, each providing specific information about the electrode's position on the scalp:

- **Letters:** These indicate the general region of the scalp where the electrode is placed:
 - **Fp:** Frontal pole
 - **F:** Frontal
 - **C:** Central
 - **P:** Parietal
 - **O:** Occipital
 - **T:** Temporal
 - **A:** Auricular (ear)
- **Odd and Even Numbers:** The numbers denote the hemisphere. Odd numbers indicate electrodes on the left hemisphere of brain and even numbers indicate electrodes on the right hemisphere of the brain.
- **Midline Electrodes:** Electrodes placed along the midline of the scalp are designated with a “z” (zero), indicating their central position. for instance: Fpz, Fz, Cz, Pz, Oz

Alternatives

While the 10-20 EEG system is widely used for its balance between coverage and ease of use, alternatives such as the 10-10 and 10-5 systems provide higher spatial resolution by increasing the number of electrodes, thereby offering more detailed brain activity mapping. High-density arrays improve source localization and are useful in research and clinical applications requiring precise brain activity data, such as epilepsy surgery planning and functional brain mapping. Additionally, specialized configurations, like customized arrays for specific cortical areas and electrocorticography (ECoG), target particular brain regions or involve direct cortical recordings, offering advantages in sensory processing studies, brain-computer interface development, and epilepsy monitoring.

2.2 Frequency Bands of EEG

EEG signals are divided into five distinct frequency bands, each associated with different types of brain activity.

Question 2.2. Determine the activities each frequency band is associated with.

- **Delta (0.5-4 Hz):** Delta waves are typically associated with deep sleep stages (slow-wave sleep) and are most prominent in infants and young children.
- **Theta (4-8 Hz):** Theta waves are linked to light sleep, relaxation, and meditative states. They are also observed during creative thinking and certain types of learning and memory processes.
- **Alpha (8-13 Hz):** Alpha waves are associated with relaxed, wakeful states, especially when the eyes are closed. They are prominent during relaxation and meditation but decrease during intense cognitive tasks or when the eyes are open.
- **Beta (13-30 Hz):** Beta waves are related to active thinking, problem-solving, and focused mental activity. They increase during states of alertness, concentration, and anxiety.
- **Gamma (30-100 Hz):** Gamma waves are associated with high-level cognitive functions, including attention, memory, and consciousness. They are involved in the integration of sensory information and the processing of complex tasks.

Understanding these frequency bands is crucial for analyzing EEG data.

2.3 Sampling Frequency and Nyquist Criterion

Question 2.3. Based on frequency bands and Nyquist criterion, which sampling frequencies are preferred for EEG signals?

In order to properly sample EEG signals according to the Nyquist criterion, the sampling frequency should be at least twice the highest frequency present in the signal. EEG signals

typically contain significant information within the frequency range of 0.1 Hz to 100 Hz. Based on this range, the preferred sampling frequencies can be determined as follows:

$$f_s \geq 2 \times f_{\max}$$

Where f_s is the sampling frequency and $f_{\max}$ is the maximum frequency of interest. For EEG signals, assuming $f_{\max} = 100$ Hz, the minimum sampling frequency should be:

$$f_s \geq 2 \times 100 \text{ Hz} = 200 \text{ Hz}$$

To ensure accurate capture of the EEG signal and allow for some margin, higher sampling frequencies are often used. Commonly preferred sampling frequencies for EEG signals include 256Hz, 512Hz and 1024Hz. These frequencies are chosen to provide a good balance between capturing the necessary detail in the EEG signals and managing data storage and processing requirements. In addition the amplitude of the EEG frequency spectrum normally drops with $1/f$ so that high frequency components have lower magnitude. This property of EEG spectrum establishes the opportunity to consider components more than 50Hz as neglectable. Following this procedure obtains EEG records with lower sampling frequency and acceptable range of error.

3 Preprocessing Data

3.1 Why Preprocessing?

Main task in phase 1 of this project is to clean given datasets following Makato's preprocessing pipeline. Preprocessing steps are essential in EEG analysis to address noise sources such as electrical interference and physiological artifacts. Techniques like filtering, artifact removal, and re-referencing isolate true neural signals and enhance data quality. These steps are crucial for accurate analysis and reliable neurophysiological insights.

3.2 Initializing EEGLAB

This project was conducted utilizing MATLAB. Specifically, the project employed EEGLAB, an extensive MATLAB toolbox designed for the analysis of electroencephalography (EEG) data. EEGLAB offers a comprehensive graphical user interface (GUI) that facilitates various stages of EEG signal processing.

For this project, two datasets were provided and subsequently loaded into EEGLAB for analysis. First dataset has 32 channels with 256Hz sampling frequency while the second dataset has 31 channels with 250Hz sampling frequency. In following sections steps of preprocessing data will be explained.

3.3 Applying Filter

In this project, a 1 Hz high-pass filter and a 50 Hz notch filter were applied to the EEG data.

The high-pass filter, with a cutoff frequency of 1 Hz, was used to remove baseline drifts, which are low-frequency variations that can obscure significant brain activity. Also an upper bound of 70Hz were applied by the filter for reducing possible high frequency noises.

Frequency responses of dataset 1 before and after applying high pass filter are shown in Figure 2 and Figure 3. (Lower bound of the filter is 1Hz, as a result, for better visualization, investigation only covers 0Hz to 3Hz) Also Frequency responses of dataset 2 in range of 0 to 3Hz before and after applying high pass filter are shown in Figure 4 and Figure 5.

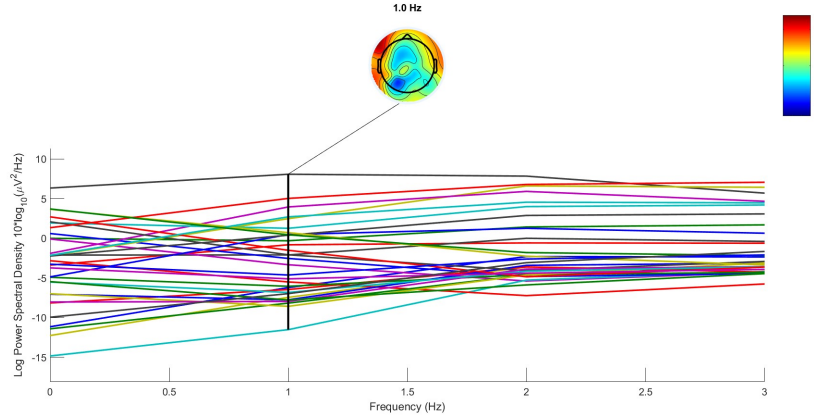


Figure 2: Spectrum of first Data Before Applying High Pass Filter

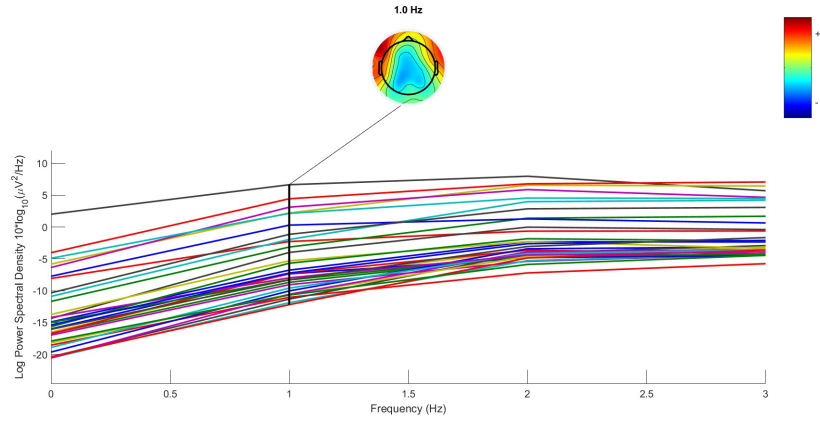


Figure 3: Spectrum of First Data After Applying High Pass Filter

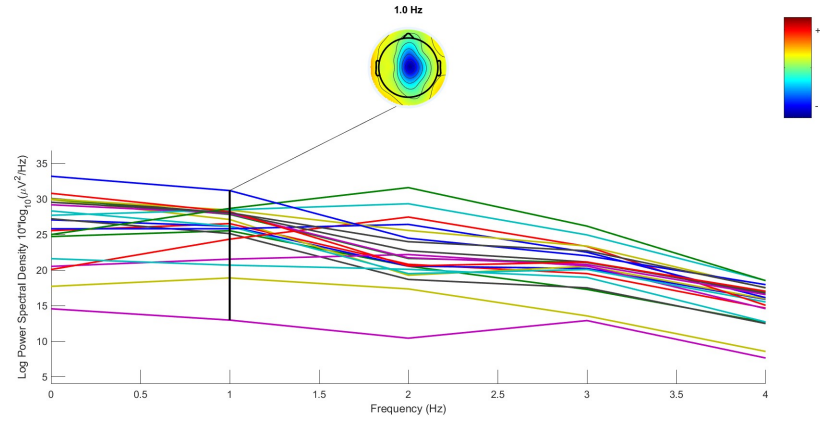


Figure 4: Spectrum of Second Data Before Applying High Pass Filter

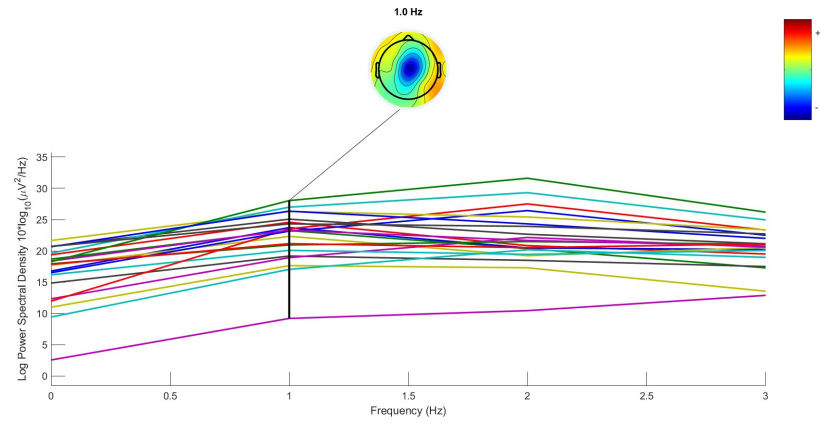


Figure 5: Spectrum of Second Data After Applying High Pass Filter

Also The 50 Hz notch filter was employed to eliminate power line noise, a common artifact in EEG recordings, which significantly enhances data quality. Notch filter was applied using the 'cleanLine' plugin which has been added to EEGLAB.

Question 3.1. Present figures showing the frequency spectrum of the Fz channel data before and after filtering.

Channel Fz spectrum of dataset 1 before and after filtering are shown in Figure 6 and Figure 7.

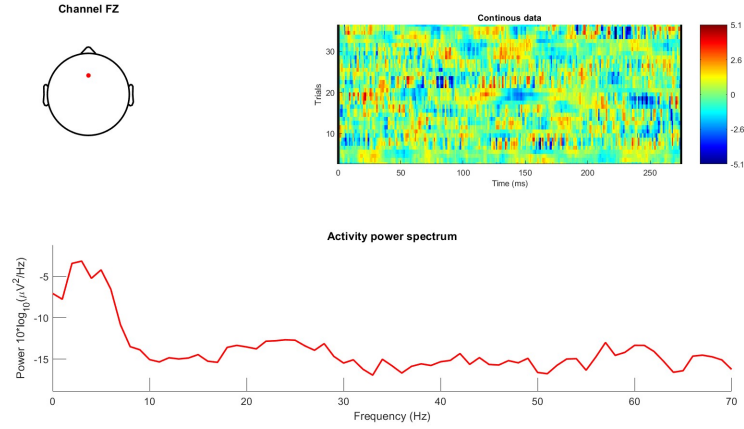


Figure 6: Dataset 1 Before Filtering

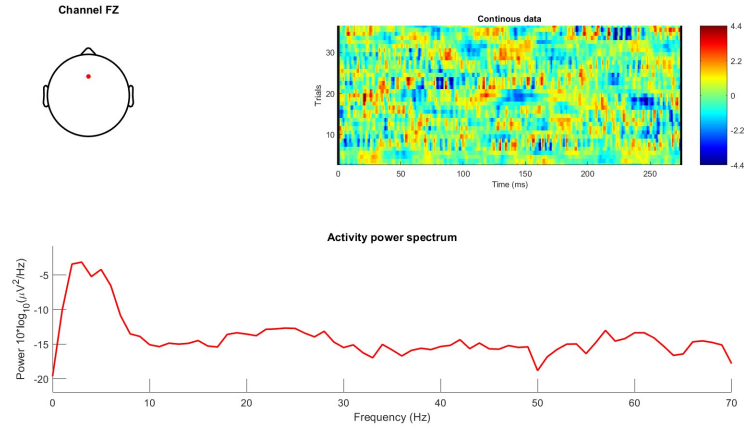


Figure 7: Dataset 1 After Filtering

Channel Fz spectrum of dataset 1 before and after filtering are shown in Figure 8 and Figure 9.

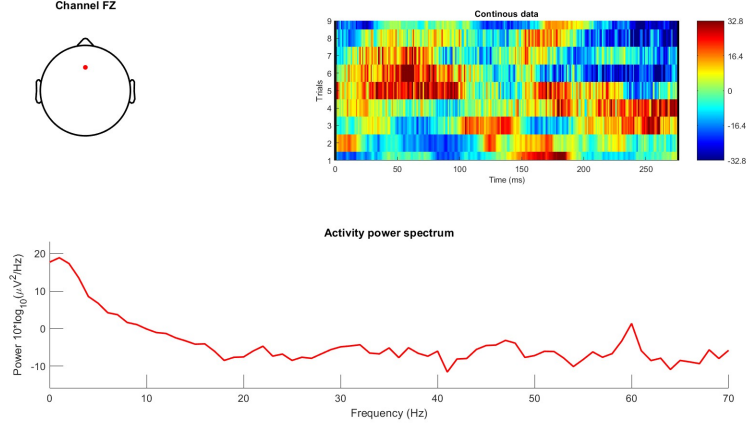


Figure 8: Dataset 2 Before Filtering

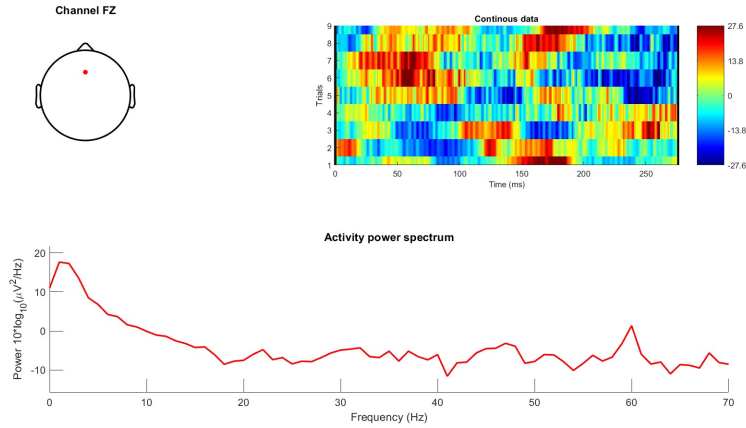


Figure 9: Dataset 2 After Filtering

3.4 Re-Referencing

Re-referencing in EEG analysis is a critical step that involves recalculating the reference electrode for each channel, often to enhance the signal quality and ensure more accurate interpretations. In this project, the average reference method was employed, where the mean of all EEG channels was used as the reference. This approach helps in reducing noise and balancing signal amplitudes across different channels, thereby improving the overall data quality. Re-referencing is particularly important after artifact removal, as it ensures that the signals from different channels are consistently referenced, which is essential for accurate and reliable interpretation of the EEG data. Additionally, re-referencing is recommended after applying filters to the data to maintain consistency in the signal reference. This process enhances the comparability of signals across different epochs and conditions, contributing to more robust and interpretable EEG analyses. Re-referencing thus plays a pivotal role in the preprocessing pipeline, facilitating the extraction of meaningful neural signals from the recorded data.

This section of preprocessing is divided into 3 parts:

- Initial Re-Referencing
- Artifact Removal
- Final Re-Referencing

In this section, we performed artifact removal, which involves the elimination of noise from each channel’s data. This process inherently affects the timing of the signals; however, the preservation of signal timing was not a critical concern for this particular project. Should the timing be of significant importance, an alternative approach can be employed. This approach entails identifying the noisy segments of the signal and reconstructing them in a manner that preserves the original timing.

3.5 Independent Component Analysis (ICA)

Independent Component Analysis (ICA) is a sophisticated technique in EEG signal processing, designed to decompose multivariate signals into additive, statistically independent components. This method is invaluable for distinguishing between neural activity and various artifacts in EEG data. Each physiological process, such as muscle contractions, produces characteristic frequency patterns that are discernible in the spectral domain. For instance, muscle activity might manifest as specific frequency peaks that can be isolated using ICA. These frequency patterns are crucial for identifying and removing artifacts.

The spatial distribution of these independent components is equally important. Understanding the origin of these components on the scalp aids in pinpointing artifact sources, which is essential for accurate EEG interpretation. ICA facilitates this by providing a spatial map of the independent components, allowing for precise localization of artifacts.

A critical application of ICA in EEG analysis is the detection of epileptic seizures. Seizure activity is often characterized by the presence of asymmetric spikes in the EEG signal. These spikes appear more prominently in certain components, indicating an abnormal neural event. By decomposing the EEG data, ICA can reveal these asymmetric spikes, which are often indicative of seizure onset.

Moreover, ICA can highlight differences in symmetric components on the scalp. For example, two components located symmetrically might exhibit different functional characteristics, with one showing specific frequency peaks absent in the other. This functional asymmetry is a hallmark of pathological conditions like seizures. The ability to detect such asymmetries allows clinicians to identify abnormal neural activities with greater precision.

Temporal analysis of the independent components further enhances the utility of ICA. By examining the time course of these components, researchers can determine the exact moments when seizure-related frequencies occur. This temporal precision is crucial for accurately identifying seizure onset and duration, which are vital for clinical diagnosis and treatment planning.

In summary, ICA is a powerful tool that enhances the accuracy and reliability of EEG analysis. By isolating independent components, ICA allows for effective artifact removal and provides detailed insights into neural activity. Its ability to detect asymmetric spikes and analyze temporal dynamics makes it particularly valuable for seizure detection and other

neurological assessments. Consequently, ICA significantly improves the overall quality of EEG signal processing and interpretation, making it an indispensable technique in both research and clinical settings.

Further analyses of ICA method and the examination of its effects on this project datasets can be found in Subsection 3.7.

3.6 Source Localization

Source localization using 'dipfit' in EEGLAB involves estimating neural source locations from EEG data through a series of methodological steps. 'Dipfit' fits equivalent dipole models to these components using a predefined head model. Accurate electrode locations and a suitable head model are crucial for precise localization. The fitted dipoles are evaluated based on residual variance. In this process, we used the MNI head model and updated the channel locations to the standard format for MNI, as the data was previously analyzed using BESA. This approach helps infer the neural sources of the observed EEG signals.

3.7 Analysis of Detected Components

Using the Independent Component Analysis (ICA) method, we identified independent components that collectively constitute the EEG signal. The components for dataset 1 are presented in Figure 10, while the components for dataset 2 are shown in Figure 11.

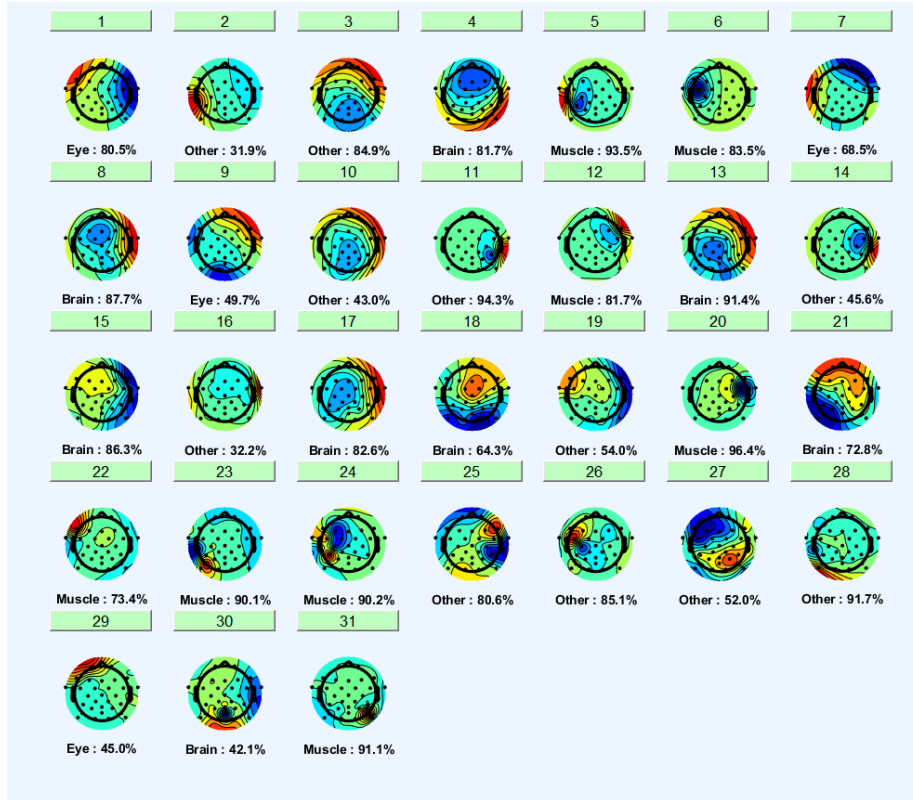


Figure 10: Sources of First EEG Signal

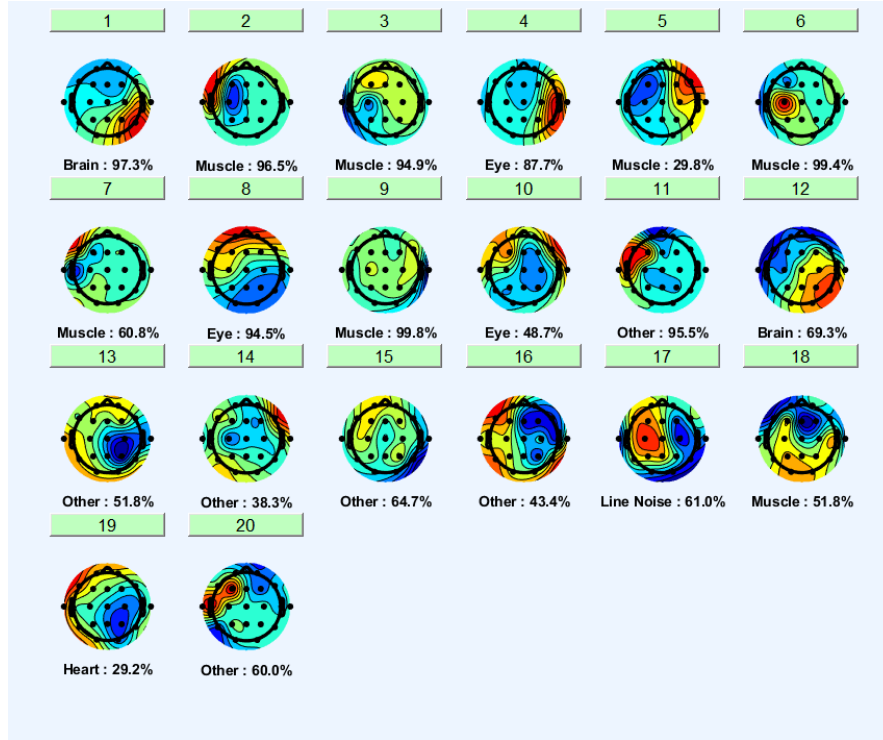


Figure 11: Sources of Second EEG Signal

These components reveal the presence of noise sources, such as eye blinks, muscle activity, and cardiac noise, which have been successfully decomposed from the initial noisy signal. This decomposition facilitates the removal of noise sources, thereby allowing for the cleaning of the data from detected noises.

Question 3.2. Provide a figure from one of the brain components identified by ICA, with details on the component.

Two components detected as brain source for first and second dataset are represented in Figure 12 and Figure 13. These two sources have been identified as brain sources with high probabilities of 91% and 97%, respectively, which is a significant finding indicating a robust identification process. The high probability values underscore the confidence in the classification of these sources as originating from brain activity. This level of certainty is particularly important in the context of EEG signal analysis, where accurate source identification can greatly enhance the interpretation of neural data.

Furthermore, an important characteristic feature of brain signal spectra is the attenuation of the frequency response amplitude at a rate of $1/f$. This spectral property is indicative of the nature of neural oscillations and their distribution across different frequencies. In the spectra of these two identified sources, this $1/f$ attenuation is clearly observable, providing additional validation for their classification as brain sources. The presence of this spectral feature suggests that these components capture genuine brain activity, distinguishing them from noise or artifact sources.

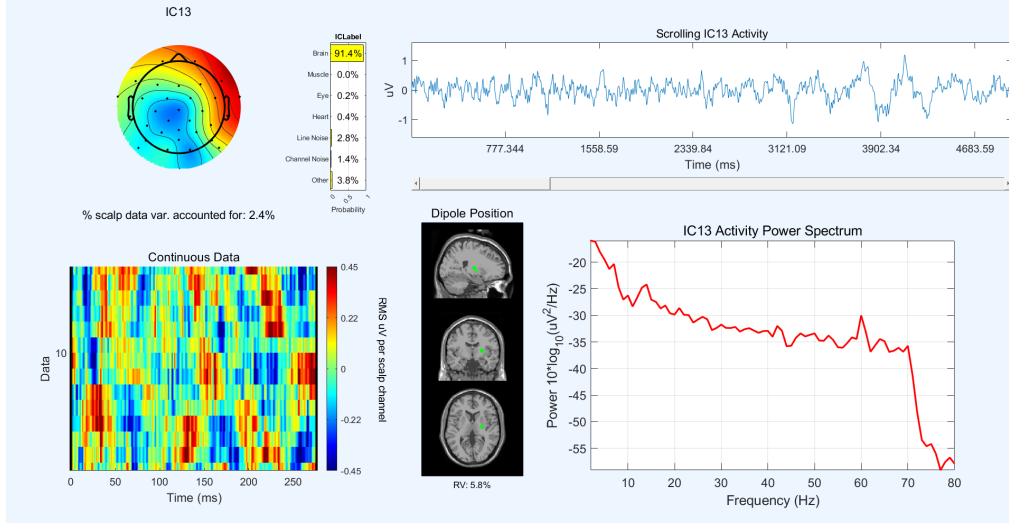


Figure 12: A Brain Source of First Signal

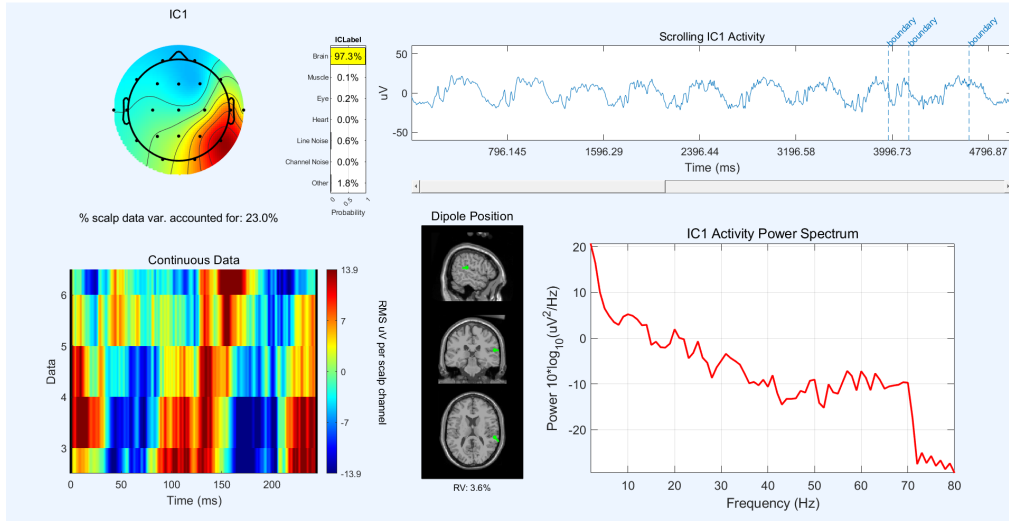


Figure 13: A Brain Source of Second Signal

Additionally, other sources exhibit distinct behaviors. For instance, one of the characteristics of muscle signals is that they have a relatively stable frequency spectrum. Compared to EEG signals, muscle signals do not show a significant amplitude decrease with increasing frequency. Also the estimated dipole position is usually very close to the surface of brain. Line noise as another possible source of noise, has high amplitude in 50Hz or 60Hz which is frequency of electrical voltage in cities.

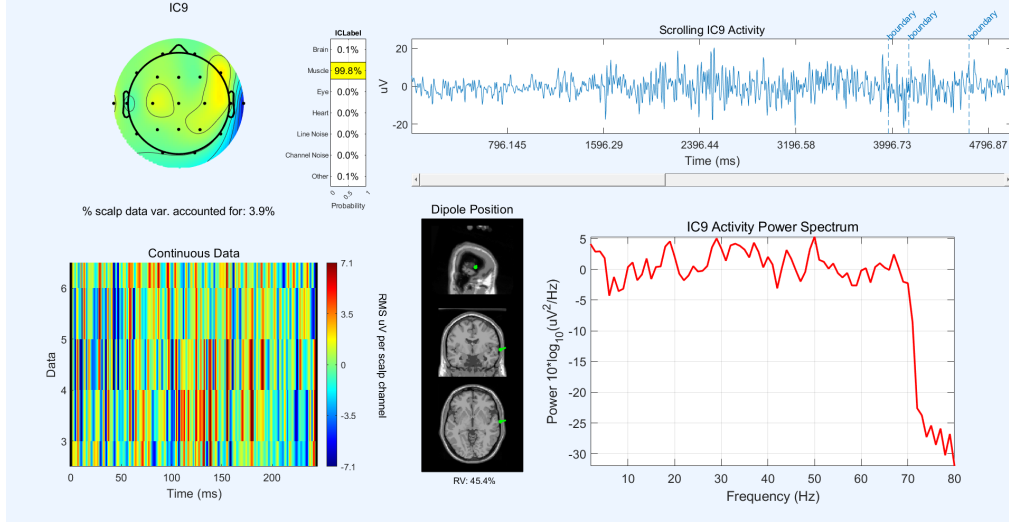


Figure 14: A Muscle Source Detected From Second EEG Signal (Amplitude Decrease After 70 Hz is Due to Filter That previously Applied)

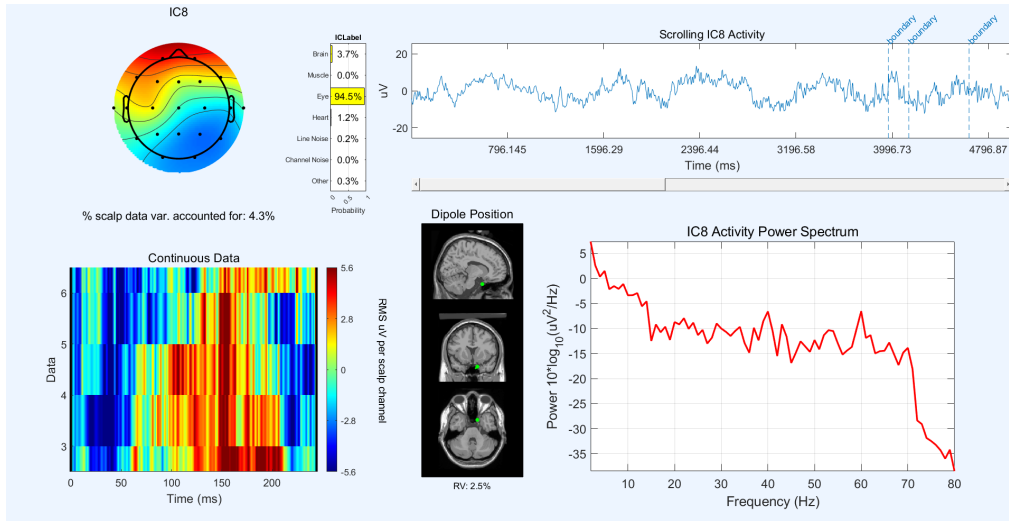


Figure 15: An Eye Source Detected From Second EEG Signal (Notice The Estimated Location of Dipole)

3.8 Epoching and Reshaping

Epoching and reshaping are crucial steps in preprocessing EEG signals to facilitate event-related analysis. Epoching involves segmenting continuous EEG data into epochs, which are specific time windows centered around events of interest. This allows for the examination of brain responses to particular stimuli or conditions. Reshaping data entails organizing these epochs into a format suitable for further analysis, ensuring consistency and ease of processing. Properly formatted and segmented data enhances the accuracy and reliability

of subsequent analyses, enabling more precise identification of event-related potentials and other neural phenomena.

3.9 Subsampling

Subsampling, specifically the selection of relevant EEG channels and the extraction of data from these channels, is a critical preprocessing step in the analysis for predicting epileptic seizures. This approach is essential for several reasons.

Firstly, selecting specific channels such as Fp1, Fp2, F3, F4, Cz, and Pz is based on their known informativeness in detecting epileptic events. These channels are strategically located in regions of the brain more likely to exhibit abnormal electrical activity associated with seizures. Focusing on these channels allows researchers to effectively identify patterns indicative of epileptic seizures, thereby increasing the likelihood of accurate detection.

Additionally, this selective approach enhances signal quality by reducing the inclusion of irrelevant or noisy data from other channels. Noise reduction results in a cleaner signal, which is crucial for identifying the subtle changes in brain activity that precede or accompany epileptic seizures. Consequently, the improved signal quality facilitates more reliable and accurate analysis.

The computational efficiency of the analysis is also significantly improved through subsampling. EEG recordings generate large volumes of data, which can be computationally intensive to process. By reducing the dataset to the most relevant channels, the computational load is decreased, allowing for faster and more efficient data processing. This efficiency is particularly important for real-time seizure prediction, where timely analysis is critical.

Furthermore, subsampling allows for a more focused analysis by concentrating on the most pertinent areas of the brain. This targeted approach increases the probability of detecting seizure-related activity, as the analysis is confined to the channels most likely to show significant changes during epileptic events. By focusing the analysis, researchers can enhance the detection capabilities and thereby improve the reliability and accuracy of seizure prediction models.